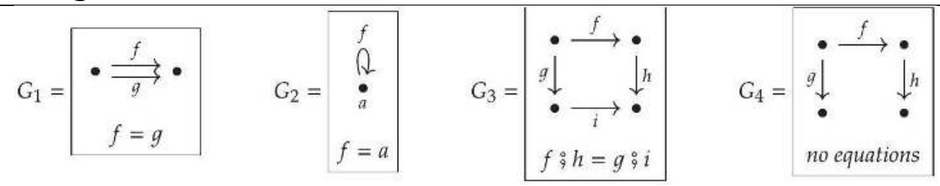
fong-spivak-invitation: residuals

equations measured 2026-09-05T11:48:20Z against the model of 2026-09-03

Unresolved

Identifier	Page	Conf.	LaTeX source	Rendered	Image
fong-spivak-invitation_ EQ0202 **[conf 0.001]**	291	■ 0.001 ● C|+70	`G_{1}=\begin{array}{|c|} \hline \text { 李 } \\ f=g \end{array} \quad G_{2}=\begin{array}{|c|} \hline f \\ \stackrel{\mathcal{Q}}{a} \\ f=a \end{array} \quad G_{3}=\begin{array}{|ccc|} \bullet & \xrightarrow{f} & \bullet \\ g \downarrow & & \downarrow \\ \bullet & i \\ f \circ & \bullet \\ f \circ g \circ i \end{array} \quad G_{4}=\begin{array}{|cc|} \bullet & \stackrel{f}{ \longrightarrow} \bullet \\ g \downarrow & \\ \bullet & \bullet \\ \text { no equations } \\ \hline \end{array}`	*(not rendered)*	
fong-spivak-invitation_ F02349	228	■ 0.185	`\operatorname{cospan} \underline{a} \xrightarrow{ \operatorname{id}_{a}} \underline{a} \stackrel{ \operatorname{id}_{\underline{a}}}{\text { 古 }}`	*(not rendered)*	(iv) identities $\mathrm{id}_a \in \mathbf{Set}(a;a)$ just given by the identity cospan $\underline{a} \xrightarrow{\mathrm{id}_{\underline{a}}} \underline{a} \xleftarrow{\mathrm{id}_{\underline{a}}} \underline{a}$.

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured

Flagged, not acted on

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>fong-spivak-invitation_
EQ0095</td><td>146</td><td>■ 0.151
● C|+37</td><td><code>\operatorname{Col}(\Phi)(a, b):= \begin{cases} \mathcal{X}(a, b) & \text { if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text { if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \varnothing & \text { if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a, b) & \text { if } a, b \in \mathcal{Y}\end{cases}</code></td><td>$\operatorname{Col}(\Phi)(a, b):= \begin{cases} \mathcal{X}(a, b) & \text { if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text { if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \varnothing & \text { if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a, b) & \text { if } a, b \in \mathcal{Y}\end{cases}$</td><td>$\mathbf{Col}(\Phi)(a, b) := \begin{cases} \mathcal{X}(a, b) & \text{if } a, b \in \mathcal{X}, \\ \Phi(a, b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y}, \\ \varnothing & \text{if } a \in \mathcal{Y}, b \in \mathcal{X}, \\ \mathcal{Y}(a, b) & \text{if } a, b \in \mathcal{Y}. \end{cases}$</td></tr>
<tr><td>fong-spivak-invitation_
EQ0134</td><td>213</td><td>■ 0.235
● C|+91</td><td><code>\begin{array}{rrrr} \operatorname{in}(\mu) & :=2, & \operatorname{in}(\eta):=0, & \operatorname{in}(\delta):=1, \end{array} \quad \operatorname{in}(\epsilon):=1,,</code></td><td>$\begin{array}{rrrr} \operatorname{in}(\mu) & :=2, & \operatorname{in}(\eta):=0, & \operatorname{in}(\delta):=1, \end{array} \quad \operatorname{in}(\epsilon):=1,,$</td><td>$\begin{array}{llll} in(\mu) := 2, & in(\eta) := 0, & in(\delta) := 1, & in(\epsilon) := 1, \\ out(\mu) := 1, & out(\eta) := 1, & out(\delta) := 2, & out(\epsilon) := 0. \end{array}$</td></tr>
<tr><td>fong-spivak-invitation_
EQ0227</td><td>306</td><td>■ 0.259
● C|+43</td><td><code>\begin{aligned} & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes\left(\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}}\right)((a, 1),(b, c, d)) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d)((b, c, d),(1, e)) \otimes \alpha((1, e), y) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} x(x, a) \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ = & x(x, y), \end{aligned}</code></td><td>$\begin{aligned} & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes\left(\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}}\right)((a, 1),(b, c, d)) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes \mathrm{U}_{\mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d)((b, c, d),(1, e)) \otimes \alpha((1, e), y) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} x(x, a) \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{\mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ = & x(x, y), \end{aligned}$</td><td>$\begin{aligned} & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})((a,1),(b,c,d)) \\ & \qquad \otimes (\epsilon_{\mathcal{X}} \times \mathrm{U}_{\mathcal{X}})((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e \in \mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d) \\ & \qquad \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e \in \mathcal{X}} \mathcal{X}(x,a) \otimes \mathcal{X}(a,b) \otimes \mathcal{X}(c,d) \otimes \mathcal{X}(b,c) \otimes \mathcal{X}(d,e) \otimes \mathcal{X}(e,y) \\ = & \mathcal{X}(x,y), \end{aligned}$</td></tr>
<tr><td>fong-spivak-invitation_
EQ0075 (3.15)</td><td>121</td><td>■ 0.380
● C|+40</td><td><code>\mathcal{G}:=\mid \stackrel{\text { Email }}{\underset{\text { received_by }}{\stackrel{\text { sent_by }}{\longrightarrow}}} \text { Address }</code></td><td>*(not rendered)*</td><td>$\mathcal{G} :=$ $\underset{\bullet}{\text{Email}} \overset{\text{sent\_by}}{\underset{\text{received\_by}}{\rightrightarrows}} \underset{\bullet}{\text{Address}}$
no equations</td></tr>
<tr><td>fong-spivak-invitation_
EQ0010 (1.7)</td><td>44</td><td>■ 0.623
● C|+21</td><td><code>p \leq g(f(p)) \quad \text { and } \quad f(g(q)) \leq q .</code></td><td>$p \leq g(f(p)) \quad \text { and } \quad f(g(q)) \leq q .$</td><td>$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$</td></tr>
<tr><td>fong-spivak-invitation_
EQ0143 (6.21)</td><td>226</td><td>■ 0.754
● C|+20</td><td><code>\begin{aligned} \circ_{i}: \mathcal{O} & \left(s_{1}, \ldots, s_{m} ; t_{i}\right) \times \mathcal{O}\left(t_{1}, \ldots, t_{n} ; t\right) \\ & \rightarrow \mathcal{O}\left(t_{1}, \ldots, t_{i-1}, s_{1}, \ldots, s_{m}, t_{i+1}, \ldots, t_{n} ; t\right) ; \end{aligned}</code></td><td>$\begin{aligned} \circ_{i}: \mathcal{O} & \left(s_{1}, \ldots, s_{m} ; t_{i}\right) \times \mathcal{O}\left(t_{1}, \ldots, t_{n} ; t\right) \\ & \rightarrow \mathcal{O}\left(t_{1}, \ldots, t_{i-1}, s_{1}, \ldots, s_{m}, t_{i+1}, \ldots, t_{n} ; t\right) ; \end{aligned}$</td><td>$\begin{aligned} \circ_i \colon & \mathcal{O}(s_1, \ldots, s_m; t_i) \times \mathcal{O}(t_1, \ldots, t_n; t) \\ & \to \mathcal{O}(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n; t); \end{aligned}$</td></tr>
<tr><td>fong-spivak-invitation_
EQ0131</td><td>206</td><td>■ 0.985
● C|+28</td><td><code>\operatorname{colim}_{\mathcal{J}} D:=\{(v, d) \mid v \in V \text { and } d \in D(v)\} / \sim</code></td><td>$\operatorname{colim}_{\mathcal{J}} D:=\{(v, d) \mid v \in V \text { and } d \in D(v)\} / \sim$</td><td>$\operatorname*{colim}_{\mathcal{J}} D := \{(v, d) \mid v \in V \text{ and } d \in D(v)\} / \sim$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
</table>

69 more flagged rows below the band, stated as a count: 40 component, 29 weak; confidence high 37, mid 12, low 20, none 0

Low confidence

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Image</th></tr>
<tr><td>fong-spivak-invitation_ EQ0135 [conf 0.019]</td><td>215</td><td>■ 0.019
●</td><td><code>\begin{aligned} \mu_{X} & :=\left(X+X \xrightarrow{\left[\mathrm{id}_{X}, \mathrm{id}_{X}\right]} X \stackrel{ \mathrm{id}_{X}}{\longleftrightarrow} X\right) \\ \eta_{X} & :=\left(\varnothing \xrightarrow{!_{X}} X \underset{\mathrm{id}_{X}}{\longleftarrow} X\right) \\ \delta_{X} & :=\left(X \xrightarrow{\mathrm{id}_{X}} X \underset{\left.\mathrm{id}_{X}, \mathrm{id}_{X} \right]}{\longrightarrow} X+X\right) \\ \epsilon_{X} & :=\left(X \xrightarrow{\mathrm{id}_{X}} \varnothing \right) \end{aligned}</code></td><td>$\mu_X := \left(X+X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \stackrel{\mathrm{id}_X}{\longleftrightarrow} X\right)$
$\eta_X := \left(\varnothing \xrightarrow{!_X} X \underset{\mathrm{id}_X}{\longleftarrow} X\right)$
$\delta_X := \left(X \xrightarrow{\mathrm{id}_X} X \underset{\mathrm{id}_X, \mathrm{id}_X]}{\longrightarrow} X+X\right)$
$\epsilon_X := \left(X \xrightarrow{\mathrm{id}_X} \varnothing\right)$</td><td>$\mu_X \coloneqq \left(X+X \xrightarrow{[\mathrm{id}_X, \mathrm{id}_X]} X \xleftarrow{\mathrm{id}_X} X\right)$
$\eta_X \coloneqq \left(\varnothing \xrightarrow{!_X} X \xleftarrow{\mathrm{id}_X} X\right)$
$\delta_X \coloneqq \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{[\mathrm{id}_X, \mathrm{id}_X]} X+X\right)$
$\epsilon_X \coloneqq \left(X \xrightarrow{\mathrm{id}_X} X \xleftarrow{!_X} \varnothing\right)$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0199 [conf 0.051]</td><td>291</td><td>■ 0.051
● S|+0</td><td><code>A, \quad A \text { : } f, \quad A \text { : } g, \quad A \text { : } f \text { : } \text { :h, } \quad A \text { : } g \text { : } i, \quad B, \quad B \text { :h, } \quad C, \quad C \text { : } i, \quad D</code></td><td>$A, \quad A : f, \quad A : g, \quad A : f \text{ :h,} \quad A : g : i, \quad B, \quad B \text{ :h,} \quad C, \quad C : i, \quad D$</td><td>$A, \quad A \fatsemi f, \quad A \fatsemi g, \quad A \fatsemi f \fatsemi h, \quad A \fatsemi g \fatsemi i, \quad B, \quad B \fatsemi h, \quad C, \quad C \fatsemi i, \quad D$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0257 (A.2) [conf 0.064]</td><td>323</td><td>■ 0.064
● S|-1</td><td><code>\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{\text { [id,id] }}{\longleftarrow} X+X \xrightarrow{\text { [id,id] }} X \stackrel{\text { id }}{\longleftarrow} X</code></td><td>$\multimap \because N \xrightarrow{\mathrm{id}} X \stackrel{\text{[id,id]}}{\longleftarrow} X+X \xrightarrow{\text{[id,id]}} X \stackrel{\text{id}}{\longleftarrow} X$</td><td> $= X \xrightarrow{\mathrm{id}} X \xleftarrow{[\mathrm{id},\mathrm{id}]} X+X \xrightarrow{[\mathrm{id},\mathrm{id}]} X \xleftarrow{\mathrm{id}} X$</td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured</td></tr>
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